Case Study: Predicting Conversion Rate 🔮

This dataset is similar to what most of the e-commerce or other companies with a website will get to work with.

Business Understanding

Optimizing conversion rate is likely the most common work of a data scientist, and rightfully so. The data revolution has a lot to do with the fact that now we can collect all sorts of data about people who buy something on our site and people who don't.

This gives us a tremendous opportunity to understand what's working well (and potentially scale it even further) and what's not working well (and fix it). The goal of this challenge is to build a model that predicts conversion rate and, based on the model, come up with ideas to improve revenue.

Business Objectives

We have data about users who hit the website: whether they converted as well as some of their characteristics such as their country, the marketing channel, their age, whether they are repeat users and the number of pages visited during that session (as a proxy for site activity/time spent on site).

Your task is to:

- Come up with recommendations for the product team and the marketing team to improve conversion rate
- Build a model to predict conversion rate

Data download

https://raw.githubusercontent.com/vkoul/data/main/misc/conversion_data.csv

Data Dictionary

There are six attributes associated with each request made by a customer:

- **country**: user country based on the IP address
- **age:** user age. Self-reported at sign-in step
- **new_user**: whether the user created the account during this session or had already an: account and simply came back to the site
- **source**: marketing channel source

    - ***Ads***: came to the site by clicking on an advertisement
    - ***Seo***: came to the site by clicking on search results
    - ***Direct***: came to the site by directly typing the URL on the browser
- **total_pages_visited**: number of total pages visited during the session. This is a proxy for time spent on site and engagement during the session.
- **converted**: this is our label. 1 means they converted within the session, 0 means they left buying nothing. The company goal is to increase conversion rate: # conversions / total sessions.